# 0. Sets and Relations

**1.** $\{\sqrt{3}, -\sqrt{3}\}$ **2.** The set is empty.

**3.** $\{1, -1, 2, -2, 3, -3, 4, -4, 5, -5, 6, -6, 10, -10, 12, -12, 15, -15, 20, -20, 30, -30, 60, -60\}$

**4.** $\{-10, -9, -8, -7, -6, -5, -4, -3, -2, -1, 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11\}$

**5.** It is not a well-defined set. (Some may argue that no element of $\mathbb{Z}^+$ is large, because every element exceeds only a finite number of other elements but is exceeded by an infinite number of other elements. Such people might claim the answer should be $\varnothing$.)

**6.** $\varnothing$ **7.** The set is $\varnothing$ because $3^3 = 27$ and $4^3 = 64$.

**8.** It is not a well-defined set. **9.** $\mathbb{Q}$

**10.** The set containing all numbers that are (positive, negative, or zero) integer multiples of 1, 1/2, or 1/3.

**11.** $\{(a,1), (a,2), (a,c), (b,1), (b,2), (b,c), (c,1), (c,2), (c,c)\}$

**12.** **a.** It is a function. It is not one-to-one since there are two pairs with second member 4. It is not onto $B$ because there is no pair with second member 2.

**b.** (Same answer as Part(**a**).)

**c.** It is not a function because there are two pairs with first member 1.

**d.** It is a function. It is one-to-one. It is onto $B$ because every element of $B$ appears as second member of some pair.

**e.** It is a function. It is not one-to-one because there are two pairs with second member 6. It is not onto $B$ because there is no pair with second member 2.

**f.** It is not a function because there are two pairs with first member 2.

**13.** Draw the line through $P$ and $x$, and let $y$ be its point of intersection with the line segment $CD$.

**14.** **a.** $\phi : [0,1] \to [0,2]$ where $\phi(x) = 2x$ **b.** $\phi : [1,3] \to [5,25]$ where $\phi(x) = 5 + 10(x-1)$
**c.** $\phi : [a,b] \to [c,d]$ where $\phi(x) = c + \frac{d-c}{b-a}(x-a)$

**15.** Let $\phi : S \to \mathbb{R}$ be defined by $\phi(x) = \tan(\pi(x - \frac{1}{2}))$.

**16.** **a.** $\varnothing$; cardinality 1 **b.** $\varnothing, \{a\}$; cardinality 2 **c.** $\varnothing, \{a\}, \{b\}, \{a,b\}$; cardinality 4
**d.** $\varnothing, \{a\}, \{b\}, \{c\}, \{a,b\}, \{a,c\}, \{b,c\}, \{a,b,c\}$; cardinality 8

**17.** **Conjecture:** $|\mathcal{P}(A)| = 2^s = 2^{|A|}$.

**Proof** The number of subsets of a set $A$ depends only on the cardinality of $A$, not on what the elements of $A$ actually are. Suppose $B = \{1, 2, 3, \cdots, s-1\}$ and $A = \{1, 2, 3, \cdots, s\}$. Then $A$ has all the elements of $B$ plus the one additional element $s$. All subsets of $B$ are also subsets of $A$; these are precisely the subsets of $A$ that do not contain $s$, so the number of subsets of $A$ not containing $s$ is $|\mathcal{P}(B)|$. Any other subset of $A$ must contain $s$, and removal of the $s$ would produce a subset of $B$. Thus the number of subsets of $A$ containing $s$ is also $|\mathcal{P}(B)|$. Because every subset of $A$ either contains $s$ or does not contain $s$ (but not both), we see that the number of subsets of $A$ is $2|\mathcal{P}(B)|$.

We have shown that if $A$ has one more element that $B$, then $|\mathcal{P}(A)| = 2|\mathcal{P}(B)|$. Now $|\mathcal{P}(\varnothing)| = 1$, so if $|A| = s$, then $|\mathcal{P}(A)| = 2^s$.

**18.** We define a one-to-one map $\phi$ of $B^A$ onto $\mathcal{P}(A)$. Let $f \in B^A$, and let $\phi(f) = \{x \in A \mid f(x) = 1\}$. Suppose $\phi(f) = \phi(g)$. Then $f(x) = 1$ if and only if $g(x) = 1$. Because the only possible values for $f(x)$ and $g(x)$ are 0 and 1, we see that $f(x) = 0$ if and only if $g(x) = 0$. Consequently $f(x) = g(x)$ for all $x \in A$ so $f = g$ and $\phi$ is one to one. To show that $\phi$ is onto $\mathcal{P}(A)$, let $S \subseteq A$, and let $h : A \to \{0, 1\}$ be defined by $h(x) = 1$ if $x \in S$ and $h(x) = 0$ otherwise. Clearly $\phi(h) = S$, showing that $\phi$ is indeed onto $\mathcal{P}(A)$.

**19.** Picking up from the hint, let $Z = \{x \in A \mid x \notin \phi(x)\}$. We claim that for any $a \in A, \phi(a) \neq Z$. Either $a \in \phi(a)$, in which case $a \notin Z$, or $a \notin \phi(a)$, in which case $a \in Z$. Thus $Z$ and $\phi(a)$ are certainly different subsets of $A$; one of them contains $a$ and the other one does not.

Based on what we just showed, we feel that the power set of $A$ has cardinality greater than $|A|$. Proceeding naively, we can start with the infinite set $\mathbb{Z}$, form its power set, then form the power set of that, and continue this process indefinitely. If there were only a finite number of infinite cardinal numbers, this process would have to terminate after a fixed finite number of steps. Since it doesn't, it appears that there must be an infinite number of different infinite cardinal numbers.

The set of everything is not logically acceptable, because the set of all subsets of the set of everything would be larger than the set of everything, which is a fallacy.

**20. a.** The set containing precisely the two elements of $A$ and the three (different) elements of $B$ is $C = \{1, 2, 3, 4, 5\}$ which has 5 elements.

i) Let $A = \{-2, -1, 0\}$ and $B = \{1, 2, 3, \cdots\} = \mathbb{Z}^+$. Then $|A| = 3$ and $|B| = \aleph_0$, and $A$ and $B$ have no elements in common. The set $C$ containing all elements in either $A$ or $B$ is $C = \{-2, -1, 0, 1, 2, 3, \cdots\}$. The map $\phi : C \to B$ defined by $\phi(x) = x + 3$ is one to one and onto $B$, so $|C| = |B| = \aleph_0$. Thus we consider $3 + \aleph_0 = \aleph_0$.

ii) Let $A = \{1, 2, 3, \cdots\}$ and $B = \{1/2, 3/2, 5/2, \cdots\}$. Then $|A| = |B| = \aleph_0$ and $A$ and $B$ have no elements in common. The set $C$ containing all elements in either $A$ of $B$ is $C = \{1/2, 1, 3/2, 2, 5/2, 3, \cdots\}$. The map $\phi : C \to A$ defined by $\phi(x) = 2x$ is one to one and onto $A$, so $|C| = |A| = \aleph_0$. Thus we consider $\aleph_0 + \aleph_0 = \aleph_0$.

**b.** We leave the plotting of the points in $A \times B$ to you. Figure 0.14 in the text, where there are $\aleph_0$ rows each having $\aleph_0$ entries, illustrates that we would consider that $\aleph_0 \cdot \aleph_0 = \aleph_0$.

**21.** There are $10^2 = 100$ numbers (.00 through .99) of the form .##, and $10^5 = 100,000$ numbers (.00000 through .99999) of the form .#####. Thus for .##### $\cdots$, we expect $10^{\aleph_0}$ sequences representing all numbers $x \in \mathbb{R}$ such that $0 \leq x \leq 1$, but a sequence trailing off in 0's may represent the same $x \in \mathbb{R}$ as a sequence trailing of in 9's. At any rate, we should have $10^{\aleph_0} \geq |[0, 1]| = |\mathbb{R}|$; see Exercise 15. On the other hand, we can represent numbers in $\mathbb{R}$ using any integer base $n > 1$, and these same $10^{\aleph_0}$ sequences using digits from 0 to 9 in base $n = 12$ would not represent all $x \in [0, 1]$, so we have $10^{\aleph_0} \leq |\mathbb{R}|$. Thus we consider the value of $10^{\aleph_0}$ to be $|\mathbb{R}|$. We could make the same argument using any other integer base $n > 1$, and thus consider $n^{\aleph_0} = |\mathbb{R}|$ for $n \in \mathbb{Z}^+, n > 1$. In particular, $12^{\aleph_0} = 2^{\aleph_0} = |\mathbb{R}|$.

**22.** $\aleph_0, |\mathbb{R}|, 2^{|\mathbb{R}|}, 2^{(2^{|\mathbb{R}|})}, 2^{(2^{(2^{|\mathbb{R}|})})}$

**23.** 1. There is only one partition $\{\{a\}\}$ of a one-element set $\{a\}$.

**24.** There are two partitions of $\{a, b\}$, namely $\{\{a, b\}\}$ and $\{\{a\}, \{b\}\}$.

**25.** There are five partitions of $\{a, b, c\}$, namely $\{\{a, b, c\}\}$, $\{\{a\}, \{b, c\}\}$, $\{\{b\}, \{a, c\}\}$, $\{\{c\}, \{a, b\}\}$, and $\{\{a\}, \{b\}, \{c\}\}$.

**26.** 15. The set $\{a, b, c, d\}$ has 1 partition into one cell, 7 partitions into two cells (four with a 1,3 split and three with a 2,2 split), 6 partitions into three cells, and 1 partition into four cells for a total of 15 partitions.

**27.** 52. The set $\{a, b, c, d, e\}$ has 1 partition into one cell, 15 into two cells, 25 into three cells, 10 into four cells, and 1 into five cells for a total of 52. (Do a combinatorics count for each possible case, such as a 1,2,2 split where there are 15 possible partitions.)

**28.** *Reflexive:* In order for $x \mathcal{R} x$ to be true, $x$ must be in the same cell of the partition as the cell that contains $x$. This is certainly true.

*Transitive:* Suppose that $x \mathcal{R} y$ and $y \mathcal{R} z$. Then $x$ is in the same cell as $y$ so $\overline{x} = \overline{y}$, and $y$ is in the same cell as $z$ so that $\overline{y} = \overline{z}$. By the transitivity of the set equality relation on the collection of cells in the partition, we see that $\overline{x} = \overline{z}$ so that $x$ is in the same cell as $z$. Consequently, $x \mathcal{R} z$.

**29.** Not an equivalence relation; 0 is not related to 0, so it is not reflexive.

**30.** Not an equivalence relation; $3 \geq 2$ but $2 \not\geq 3$, so it is not symmetric.

**31.** It is an equivalence relation; $\overline{0} = \{0\}$ and $\overline{a} = \{a, -a\}$ for $a \in \mathbb{R}, a \neq 0$.

**32.** It is not an equivalence relation; $1 \mathcal{R} 3$ and $3 \mathcal{R} 5$ but we do not have $1 \mathcal{R} 5$ because $|1 - 5| = 4 > 3$.

**33.** (See the answer in the text.)

**34.** It is an equivalence relation;

$$\overline{1} = \{1, 11, 21, 31, \cdots\}, \quad \overline{2} = \{2, 12, 22, 32, \cdots\}, \quad \cdots, \quad \overline{10} = \{10, 20, 30, 40, \cdots\}.$$

**35.** (See the answer in the text.)

**36.** **a.** Let $h, k,$ and $m$ be positive integers. We check the three criteria.
*Reflexive:* $h - h = n0$ so $h \sim h$.
*Symmetric:* If $h \sim k$ so that $h - k = ns$ for some $s \in \mathbb{Z}$, then $k - h = n(-s)$ so $k \sim h$.
*Transitive:* If $h \sim k$ and $k \sim m$, then for some $s, t \in \mathbb{Z}$, we have $h - k = ns$ and $k - m = nt$. Then $h - m = (h - k) + (k - m) = ns + nt = n(s + t)$, so $h \sim m$.

**b.** Let $h, k \in \mathbb{Z}^+$. In the sense of this exercise, $h \sim k$ if and only if $h - k = nq$ for some $q \in \mathbb{Z}$. In the sense of Example 0.19, $h \equiv k \pmod{n}$ if and only if $h$ and $k$ have the same remainder when divided by $n$. Write $h = nq_1 + r_1$ and $k = nq_2 + r_2$ where $0 \leq r_1 < n$ and $0 \leq r_2 < n$. Then

$$h - k = n(q_1 - q_2) + (r_1 - r_2)$$

and we see that $h - k$ is a multiple of $n$ if and only if $r_1 = r_2$. Thus the conditions are the same.

**c.** **a.** $\overline{0} = \{\cdots, -2, 0, 2, \cdots\}, \quad \overline{1} = \{\cdots, -3, -1, 1, 3, \cdots\}$

**b.** $\overline{0} = \{\cdots, -3, 0, 3, \cdots\}, \quad \overline{1} = \{\cdots, -5, -2, 1, 4, \cdots\}, \quad \overline{2} = \{\cdots, -1, 2, 5, \cdots\}$

**c.** $\overline{0} = \{\cdots, -5, 0, 5, \cdots\}, \quad \overline{1} = \{\cdots, -9, -4, 1, 6, \cdots\}, \quad \overline{2} = \{\cdots, -3, 2, 7, \cdots\},$
$\overline{3} = \{\cdots, -7, -2, 3, 8, \cdots\}, \quad \overline{4} = \{\cdots, -1, 4, 9, \cdots\}$